

# Engineering Mathematics Study Notes: Limit Comparison Test Solved Problems

## 1. Concept Identification

The notes apply the **Limit Comparison Test (LCT)** combined with the **P-Series Test** to advanced algebraic series containing fractional factors and radical terms. It also introduces **Rationalization** as a preliminary mechanical step to convert difference-of-roots expressions into fraction forms suitable for testing.

---

## 2. Mathematical Representations

### Difference of Squares Identity (Rationalization)

To remove radical terms subtraction from the numerator, apply the conjugate identity:

$$(a - b)(a + b) = a^2 - b^2$$

### Limit Comparison Test Ratio

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = L \quad (0 < L < \infty)$$

---

## 3. The Algorithm (Procedural Recipe)

To solve any complex fractional or radical infinite series from first principles, follow this exact procedural blueprint:

### Step 1: Pre-Process with Rationalization (If Radicals are Subtracted)

If  $u_n$  consists of two square roots subtracted from each other, multiply the numerator and denominator by its conjugate counterpart (changing the minus sign to a plus sign) to force a fractional form.

### Step 2: Establish the Target Term ( $u_n$ )

Isolate the simplified algebraic expression inside the summation sign.

### Step 3: Determine the Helper Term ( $v_n$ )

Extract the single highest order term of  $n$  from the numerator and divide it by the single highest order term of  $n$  from the denominator. Simplify this down to a pure p-series structure:

$$v_n = \frac{1}{n^p}$$

### Step 4: Evaluate the Limit of the Ratio ( $\frac{u_n}{v_n}$ )

Compute the limit as  $n \rightarrow \infty$  of the product of  $u_n$  and the mathematical inverse of  $v_n$ . Factor out leading degrees of  $n$ , eliminate terms approaching zero, and solve for a finite constant value  $L$ .

## Step 5: Execute the Benchmark Verification

Check  $p$  against the p-series condition ( $p > 1$  converges,  $p \leq 1$  diverges) to determine the fate of  $\sum v_n$ . Conclude that  $\sum u_n$  shares the same outcome.

---

## 4. Operational Mechanics & Solutions to Board Problems

### Solution to Problem 1: Fractional Factored Series

The lecture board presents the following problem:

$$\text{Test for convergence of the series: } \frac{1}{1 \cdot 2 \cdot 3} + \frac{3}{2 \cdot 3 \cdot 4} + \frac{5}{3 \cdot 4 \cdot 5} + \dots \infty$$

#### Step 1: Construct the general term formula ( $u_n$ )

Examine the pattern of the numerators: 1, 3, 5, ... which forms an arithmetic progression where  $a = 1, d = 2$ :

$$\text{Top Term} = 1 + (n - 1)2 = 2n - 1$$

Examine the pattern of the three multiplying numbers in the denominators:

$$\text{First factors: } 1, 2, 3 \dots \Rightarrow n$$

$$\text{Second factors: } 2, 3, 4 \dots \Rightarrow (n + 1)$$

$$\text{Third factors: } 3, 4, 5 \dots \Rightarrow (n + 2)$$

Combine these together to form the target expression:

$$u_n = \frac{2n - 1}{n(n + 1)(n + 2)}$$

#### Step 2: Isolate the maximum powers to build $v_n$

The highest power of  $n$  visible in the numerator is  $n^1$ .

Multiply out the leading terms of the denominator to find its maximum power degree:

$$n \times n \times n = n^3$$

Form the comparison helper term using these maximum degrees:

$$v_n = \frac{n}{n^3} = \frac{1}{n^2}$$

#### Step 3: Setup the ratio calculation

Set up the limit function by multiplying  $u_n$  by the reciprocal of  $v_n$ :

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \left[ \frac{2n - 1}{n(n + 1)(n + 2)} \times \frac{n^2}{1} \right]$$

Multiply out the expressions:

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{2n^3 - n^2}{n^3 + 3n^2 + 2n}$$

Forcefully pull out the maximum global power ( $n^3$ ) from all positions:

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{n^3 \left(2 - \frac{1}{n}\right)}{n^3 \left(1 + \frac{3}{n} + \frac{2}{n^2}\right)}$$

Cancel the  $n^3$  elements outside the brackets:

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{2 - \frac{1}{n}}{1 + \frac{3}{n} + \frac{2}{n^2}}$$

Substitute  $n = \infty$ , turning all inverse elements to 0:

$$L = \frac{2 - 0}{1 + 0 + 0} = 2$$

Since 2 is a real, finite, non-zero constant, the Limit Comparison Test is active and valid.

#### Step 4: Run the p-series assessment

Look at the helper term exponent from Step 2:

$$v_n = \frac{1}{n^2} \implies p = 2$$

Because  $2 > 1$ , the helper series  $\sum v_n$  is convergent.

#### Step 5: Final Conclusion

By the rules of the Limit Comparison Test, the original target series is **Convergent**.

## Solution to Problem 2: Radical Difference Series

The lecture board presents the following radical problem:

$$\text{Test the convergence of the series: } \sum_{n=1}^{\infty} (\sqrt{n+1} - \sqrt{n-1})$$

#### Step 1: Rationalize the expression to create a fraction

The target expression is:

$$u_n = \sqrt{n+1} - \sqrt{n-1}$$

Multiply both the top and hidden bottom by the conjugate addition expression:

$$u_n = \frac{(\sqrt{n+1} - \sqrt{n-1}) \times (\sqrt{n+1} + \sqrt{n-1})}{\sqrt{n+1} + \sqrt{n-1}}$$

Apply the algebraic difference of squares identity  $(a - b)(a + b) = a^2 - b^2$  to collapse the top numerator:

$$u_n = \frac{(\sqrt{n+1})^2 - (\sqrt{n-1})^2}{\sqrt{n+1} + \sqrt{n-1}}$$

Extract values out of the canceled square roots:

$$u_n = \frac{(n+1) - (n-1)}{\sqrt{n+1} + \sqrt{n-1}}$$

Distribute the negative sign carefully:

$$u_n = \frac{n+1 - n+1}{\sqrt{n+1} + \sqrt{n-1}}$$

$$u_n = \frac{2}{\sqrt{n+1} + \sqrt{n-1}}$$

#### Step 2: Formulate the comparison helper term ( $v_n$ )

The numerator contains no variable, so its highest degree is  $n^0 = 1$ .

The denominator's dominant terms are locked inside square roots, which means their highest power degree is  $\sqrt{n} = n^{1/2}$ .

$$v_n = \frac{1}{\sqrt{n}} = \frac{1}{n^{1/2}}$$

**Step 3: Evaluate the limit value**

Multiply  $u_n$  by the mathematical reciprocal of  $v_n$ :

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \left[ \frac{2}{\sqrt{n+1} + \sqrt{n-1}} \times \frac{\sqrt{n}}{1} \right]$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{2\sqrt{n}}{\sqrt{n} \left( \sqrt{1 + \frac{1}{n}} + \sqrt{1 - \frac{1}{n}} \right)}$$

Cancel the matching  $\sqrt{n}$  terms from both positions:

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{2}{\sqrt{1 + \frac{1}{n}} + \sqrt{1 - \frac{1}{n}}}$$

Apply the infinite limit to eliminate the internal fractional values:

$$L = \frac{2}{\sqrt{1+0} + \sqrt{1-0}} = \frac{2}{1+1} = \frac{2}{2} = 1$$

Since 1 is a finite, non-zero constant, the test framework confirms identical behavior.

**Step 4: Execute the p-series test**

Check the value of the exponent from Step 2:

$$p = \frac{1}{2}$$

Evaluate the threshold condition:

$$\frac{1}{2} \leq 1 \implies \text{The helper series } \sum v_n \text{ is Divergent.}$$

**Step 5: Final Conclusion**

By the mechanical rules of the Limit Comparison Test, the original target radical series is **Divergent**.